Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf 2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? _____________ S0/0/0 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? __________ 129 2

Does R2 show up as an OSPF neighbor on R1? _________ Yes 2

Does R2 show up as an OSPF neighbor on R3? _________ No 2

What does this information tell you?

Every traffic from R3 to 192.168.2.0/24 network need to be routed by R1.

The R2 S0/0/0 interface is still inactive, so the OSPF information can not be advertised through this interface.

2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config) # router ospf 1

R2(config-router) # no passive-interface s0/0/1

2

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? ____________ s0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

Accumulated cost:65

1 serial link(64 cost) + R2 G 0/0 link(1 cost)

Is R2 listed as an OSPF neighbor to R3? ________ yes

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

Since the most of current link are faster than 100mb/s. It's necessary to use higher default reference-bandwidth to calculate the cost of these links.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

Cost to 192.168.3.0/24: R1 S0/0/1(781 cost) + R3 G0/0 (1cost) =782

Cost to 192.168.23.0/30: R1 S0/0/1(781 cost) + R3 S0/0/1(64 cost) = 845

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

1562. The route to 192.168.23.0/24 go through 2 serial links, 2x781=1562

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

ODPF will select the shortest path. From R1 s0/0/0 to R2 s0/0/, then R3 G0/0 , the cost is 1563, which is smaller than the cost, from R1 so/o/1 directly connect to R3 G0/0, is 1566

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Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

It is used in DR/BDR election and as the originator of routes. If not controlled, it may be automatically selected based on the highest IP address of loopback or physical interfaces, leading to unpredictable changes if interfaces go down. Controlling it ensures stability and predictability in the network.

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2. Why is the DR/BDR election process not a concern in this lab?

DR/BDR election process is a problem in multi-access network. The link in this lab are point to point links.

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3. Why would you want to set an OSPF interface to passive?

For interface connected to network where OSPF does not exist, it is helpful to reduce unnecessary traffic.

4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

OSPF's hierarchical design is achieved through the use of areas. By dividing an autonomous system into multiple areas, OSPF limits the flooding of link-state updates to within each area, reducing routing overhead.

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5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

Network 10.0.0.0 0.255.255.255 area 0

4

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

OSPF will be used, since it has the shorter path compare to RIP and EIGRP.

0

EIGRP